

# Dark Energy as Residual Vacuum Free Energy: A Thermodynamic Bound on the Cosmological Constant

*Part of the FST Program [35]*

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## Abstract

This revision audits the mathematical core of the proposed thermodynamic dark-energy framework. For the dimensionally closed reduced function  $\Phi_{\text{red}} = A + B\rho \ln(\rho/\rho_{\text{Pl}}) + C\rho^2$ ,  $B, C > 0$ , existence, uniqueness, strict convexity, and an explicit Lambert- $W$  minimizer are proven. The original momentum-space expression is dimensionally inhomogeneous and does not determine the matching coefficients. Consequently, the scaling  $\Lambda_{\text{eff}} \sim H_0^2/[c^2 \ln(M_{\text{Pl}}/H_0)]$  remains an external RG-matching ansatz. Its corrected numerical value is  $3.8 \times 10^{-55} \text{ m}^{-2}$ , about 290 times below observation.

The scalar-sector audit also changes the physical status. The Pöschl–Teller potential  $V_0 \text{sech}^2(\phi/\phi_0)$  has an unstable maximum at  $\phi = 0$ ; the earlier tanh history is a phenomenological proxy, not a derived stable attractor. Standard quasistatic metric  $f(R)$  perturbations approach a slip of  $1/2$  (or reciprocal 2, by convention) and a GR-like lensing combination  $\Sigma \simeq 1/F$ , not  $5/4$  and  $9/8$ . Finally, the Hu–Sawicki environment ledger yields correct necessary curvature thresholds but no source-bound local profile, so neither  $|f_{R0}| = 1.2 \times 10^{-5}$  nor  $10^{-6}$  is by itself a Cassini proof. The framework is therefore a conditional research program, not a closed solution of the cosmological-constant or screening problems. **Keywords:** cosmological constant,

dark energy, vacuum energy, scalar-tensor gravity, gravitational slip

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# 1 Introduction

This revision records the mathematical status after a line-by-line proof/paper audit on 16 July 2026. The purpose is narrower than that of the earlier skeleton: it separates the statements that follow from a dimensionally closed one-variable model from the physical identifications that remain open.

The audit changes four headline claims. First, the original momentum-space integrand was dimensionally inhomogeneous and therefore did not define a physical free-energy density without explicit matching coefficients. Second, the proposed Pöschl–Teller saturation law was not derived from the canonical Klein–Gordon–Friedmann system and its hilltop is unstable. Third, the thermodynamic beta function defined as the derivative of a stationarity condition is identically zero. Fourth, standard metric  $f(R)$  perturbation theory does not predict the previously stated values  $\eta = 5/4$  and  $\Sigma = 9/8$ . These claims are withdrawn below. Table 1 details the evidentiary gates for each sector, and Figure 1 provides an architectural schematic of the resulting four-tier epistemic hierarchy.

Table 1: Claim-Status and Evidence-Gate Ledger for the Present Revision.

| Layer                          | What is established here                                                                           | Stop gate before a stronger claim                                                                       |
|--------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Reduced variational core       | Existence, uniqueness, strict convexity, and the Lambert- $W$ minimizer for specified $B, C > 0$ . | Independent physical derivation of $B(H)$ , $C(H)$ , and the subtraction scheme.                        |
| Conditional cosmological scale | The corrected ansatz value is $3.8 \times 10^{-55} \text{ m}^{-2}$ , about 290 below observation.  | No target-value input when fixing $C/B$ ; compare only after the RG ledger is closed.                   |
| Scalar and perturbation sector | Earlier hilltop, beta-response, $\eta = 5/4$ , and $\Sigma = 9/8$ claims are withdrawn.            | Stable scalar history and standard $f(R)$ response must be derived from the full equations.             |
| Local and observational gates  | Environment thresholds are arithmetically defined; proxy MCMC is background-level only.            | Source-bound local $f_R(\text{env})$ profile and official Planck/DESI likelihood.                       |
| RVM/entropy audit              | The 3 August 2026 evaluator is only an exploratory insertion calculation.                          | Derive $\nu$ , horizon entropy, $A(E)$ , ensemble, and Cassini convention without inserted pass values. |

## 2 Dimensionally Closed Variational Model

### 2.1 Raw Motivation and Its Limitations

The earlier motivating expression was

$$\Phi_{\text{raw}}(\rho) = \int_{\Lambda_{\text{IR}}}^{\Lambda_{\text{UV}}} \left[ \frac{\omega(k)}{2} + T_{\text{dS}} \rho \ln \left( \frac{\rho}{\rho_{\text{Pl}}} \right) + \frac{8\pi G \rho^2}{k^2} \right] \frac{4\pi k^2}{(2\pi)^3} dk. \quad (1)$$

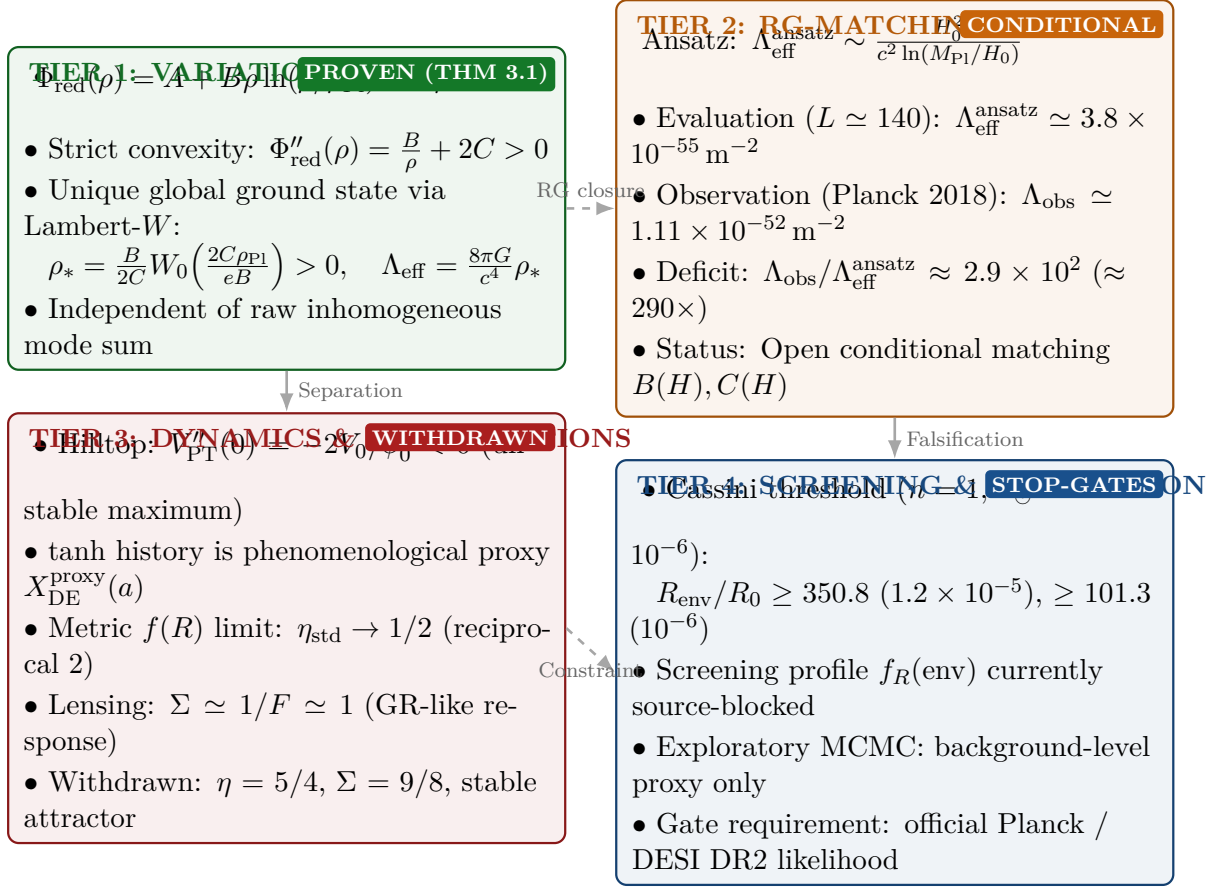


Figure 1: Epistemic architecture and falsification gate structure of the FST dark-energy framework. The mathematically rigorous variational core (Tier 1, green) is strictly decoupled from the conditional RG-matching ansatz (Tier 2, orange), the corrected/withdrawn dynamical and perturbation claims (Tier 3, red), and the unfulfilled local screening and official likelihood gates (Tier 4, blue).

In natural units, the three terms inside the square brackets have dimensions  $E$ ,  $E^5$ , and  $E^4$ , respectively. They cannot be added as written. Planck-unit rescaling does not repair the problem: it hides powers of  $M_{\text{Pl}}$  and  $H$ , precisely the powers that determine the proposed cosmological scaling. Equation (1) is therefore only a schematic motivation, not a physical functional and not a derivation from a mode sum.

## 2.2 Reduced Model

**Definition 2.1** (Reduced Free-Energy Density). For  $\rho > 0$ , let

$$\Phi_{\text{red}}(\rho) = A + B\rho \ln\left(\frac{\rho}{\rho_{\text{Pl}}}\right) + C\rho^2, \quad B > 0, \quad C > 0. \quad (2)$$

Here  $[\rho] = [\Phi_{\text{red}}] = E^4$ ,  $[A] = E^4$ ,  $[B] = 1$ , and  $[C] = E^{-4}$ . The constants  $B$  and  $C$  are renormalized matching coefficients. Their values are not supplied by Eq. (1).

The kernel

$$V_{\text{grav}}(\rho, k) = \frac{8\pi G\rho^2}{k^2} \quad (3)$$

has the dimension of an energy density and can motivate a positive quadratic term after an explicitly specified integration and matching prescription. Dimensional analysis selects it as a lowest-order monomial only after the assumptions of one power of  $G$ , quadratic dependence on  $\rho$ , and inverse powers of  $k$  are imposed. It does not establish uniqueness among general kernels, and a quasistatic 4/3 force enhancement does not by itself derive an off-shell free-energy functional.

### 3 Proven Result and Conditional Scaling

**Theorem 3.1** (Unique Minimizer of the Reduced Model). *The function  $\Phi_{\text{red}}$  in Definition 2.1 has exactly one global minimizer  $\rho_* > 0$ . It satisfies*

$$B \left[ 1 + \ln \left( \frac{\rho_*}{\rho_{\text{Pl}}} \right) \right] + 2C\rho_* = 0 \quad (4)$$

and is given explicitly by

$$\rho_* = \frac{B}{2C} W \left( \frac{2C\rho_{\text{Pl}}}{eB} \right), \quad (5)$$

where  $W$  is the principal Lambert function.

*Proof.* The first derivative tends to  $-\infty$  as  $\rho \rightarrow 0^+$  and to  $+\infty$  as  $\rho \rightarrow \infty$ . Moreover,

$$\Phi''_{\text{red}}(\rho) = \frac{B}{\rho} + 2C > 0. \quad (6)$$

Thus the derivative has exactly one zero and strict convexity makes that zero the unique global minimizer. Rearranging Eq. (4) gives  $(2C\rho_*/B) \exp(2C\rho_*/B) = 2C\rho_{\text{Pl}}/(eB)$ , which yields Eq. (5).  $\square$

The associated curvature convention is

$$\Lambda_{\text{eff}} = \frac{8\pi G}{c^4} \rho_*. \quad (7)$$

No dependence on  $H_0$  follows until  $B/C$  is independently matched. The frequently used expression

$$\Lambda_{\text{eff}}^{\text{ansatz}} \sim \frac{H_0^2}{c^2 \ln(M_{\text{Pl}}/H_0)} \quad (8)$$

is therefore a conditional RG-matching ansatz, not a consequence of Theorem 3.1.

**Corollary 3.2** (Arithmetic of the Conditional Ansatz). *For  $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $L = \ln(M_{\text{Pl}}/H_0) \simeq 140$ ,*

$$\frac{H_0^2}{c^2 L} \simeq 3.8 \times 10^{-55} \text{ m}^{-2}. \quad (9)$$

*Compared with  $\Lambda_{\text{obs}} \simeq 1.11 \times 10^{-52} \text{ m}^{-2}$  from Planck 2018 [4], the ansatz is low by a factor of about  $2.9 \times 10^2$ . This is an unresolved normalization/matching factor, not order-of-magnitude agreement.*

**Physical-identification gate.** The curvature convention in Eq. (7) is conditional on an independent identification of the model parameter with the gravitating energy density, including its conservation or exchange law and the subtraction term  $A$ . No external RVM or horizon result fixes these choices or removes the factor-290 matching deficit automatically.

### 3.1 Scalar-Sector Status

Consider the phenomenological Lagrangian

$$\mathcal{L} = \frac{R + \gamma R^2}{16\pi G} + \frac{1}{2}(\partial\phi)^2 - V_{\text{PT}}(\phi) + \mathcal{L}_m, \quad V_{\text{PT}}(\phi) = V_0 \text{sech}^2(\phi/\phi_0). \quad (10)$$

For the quadratic curvature sector the scalaron mass is  $m_s^2 = 1/(6\gamma)$ . The scalar potential obeys

$$V'_{\text{PT}}(0) = 0, \quad V''_{\text{PT}}(0) = -\frac{2V_0}{\phi_0^2} < 0. \quad (11)$$

Hence  $\phi = 0$  is a maximum, not a minimum. Linear perturbations satisfy

$$r^2 + 3Hr - \frac{2V_0}{\phi_0^2} = 0,$$

and the larger root is positive for every  $V_0 > 0$ . Hubble friction can delay the roll but cannot make the hilltop asymptotically stable. The earlier stable de Sitter theorem and oscillation forecast are withdrawn.

For a canonical scalar and pressureless matter, define

$$x = \frac{\dot{\phi}}{\sqrt{6} H M_{\text{Pl}}}, \quad y = \frac{\sqrt{V}}{\sqrt{3} H M_{\text{Pl}}}, \quad \lambda = -M_{\text{Pl}} \frac{V'}{V}, \quad N = \ln a.$$

The correct autonomous system is

$$x' = -3x + \sqrt{\frac{3}{2}} \lambda y^2 + \frac{3}{2} x(1 + x^2 - y^2), \quad (12)$$

$$y' = -\sqrt{\frac{3}{2}} \lambda x y + \frac{3}{2} y(1 + x^2 - y^2). \quad (13)$$

With  $\Omega_\phi = x^2 + y^2$  and  $w_\phi = (x^2 - y^2)/(x^2 + y^2)$ ,

$$\frac{d\Omega_\phi}{dN} = -3w_\phi \Omega_\phi (1 - \Omega_\phi). \quad (14)$$

This does not reduce to the tanh equation printed in earlier versions.

A bounded tanh curve may still be used as a data parameterization if it is labeled as such. A dimensionally consistent non-negative choice is

$$X_{\text{DE}}^{\text{proxy}}(a) = \frac{\Phi_0}{2} \left[ 1 + \tanh(\kappa(a - a_t)) \right], \quad (15)$$

$$\frac{dX_{\text{DE}}^{\text{proxy}}}{da} = 2\kappa X_{\text{DE}}^{\text{proxy}} \left( 1 - \frac{X_{\text{DE}}^{\text{proxy}}}{\Phi_0} \right). \quad (16)$$

Here  $X_{\text{DE}} = \rho_{\text{DE}}/\rho_{\text{crit},0}$ , not the instantaneous fraction  $\rho_{\text{DE}}/\rho_{\text{crit}}(a)$ . Equations (15)–(16) are a phenomenological proxy and are not derived from Eq. (10).

## 4 Kinematic Consequences of a Density Proxy

The conditional estimate from Eq. (8) is  $3.8 \times 10^{-55} \text{ m}^{-2}$ , not  $3.8 \times 10^{-53} \text{ m}^{-2}$ . It is therefore about 290 times below  $\Lambda_{\text{obs}}$ , not a factor of three.

A bounded curve may be used as a phenomenological density history only if its normalization is defined consistently. Let  $X_{\text{DE}}(a) = \rho_{\text{DE}}(a)/\rho_{\text{crit},0}$  and

$$E^2(a) = \frac{H^2}{H_0^2} = \Omega_m a^{-3} + X_{\text{DE}}(a).$$

Then

$$q(a) = -1 - \frac{a}{2E^2} \frac{dE^2}{da} = \frac{\frac{1}{2}\Omega_m a^{-3} - X_{\text{DE}} - \frac{a}{2}X'_{\text{DE}}}{E^2}, \quad (17)$$

$$w_{\text{DE}}(a) = -1 - \frac{1}{3} \frac{d \ln X_{\text{DE}}}{d \ln a}. \quad (18)$$

The earlier deceleration formula used the wrong coefficient for  $aX'_{\text{DE}}$ . It also alternated between an instantaneous density fraction and a density normalized to today's critical density. Consequently, the reported acceleration redshift,  $w_0$ , CPL map, and proxy  $\chi^2$  are withdrawn pending recomputation from a physically derived history. No official DESI likelihood evaluation is claimed.

*Remark 4.1* (DESI DR2 Comparators and the Open Model-Identification Gate). Scalar-tensor reconstructions [36] and interacting quintessence with field-dependent dark-matter masses [37] provide distinct mechanisms for an effective phantom crossing. The quintom model of [38] instead includes a canonical and a phantom field; it is not the same ghost-free interacting-quintessence mechanism. These external models do not identify the reduced minimizer with a conserved gravitational energy density or close its matching, exchange, horizon/ensemble, local-screening, or likelihood gates. The tanh proxy alone establishes no phantom-crossing mechanism.

## 5 Local Consistency Gates

### 5.1 Radiation-Era Trace

For an ideal perfect fluid with the sign convention used here,

$$T = (1 - 3w)\rho.$$

Thus  $T = 0$  for exactly massless radiation with  $w = 1/3$ . This removes the leading classical trace source, but it is not a proof of BBN safety: massive species, trace anomalies, background evolution, and initial conditions must be propagated through a quantitative abundance calculation.

### 5.2 Convexity and Screening

A fixed strictly convex function of  $\rho$  has one minimizer. This elementary fact does not constitute a no-go theorem for chameleon screening. In a chameleon model the effective scalar potential depends on ambient density, so the function being minimized changes with the environment. The previous  $\Gamma$ -convergence and energy-dissipation block did not supply

a valid liminf/recovery proof and mixed gradients of different variables; it is withdrawn. The accompanying numerical phase plot remains, at most, a toy illustration and not a Cassini calculation.

### 5.3 Scope of the Variational Statement

The variational theorem below concerns only the reduced scalar function with specified positive coefficients. It does not solve the radiative-stability problem, determine a vacuum subtraction scheme, construct an RG trajectory, or derive a local screening profile. Those are independent gates. In particular, neither a resolvent analogy nor a metric on configuration space changes the zero set of the differential of a fixed scalar functional.

## 6 Renormalization, Beta Response, and Observational Status

The physical scaling problem is concentrated in the matching coefficients  $B$  and  $C$ . A valid derivation must provide a source- and scheme-controlled ledger

bare  $\longrightarrow$  Minkowski subtraction  $\longrightarrow$  counterterms  $\longrightarrow$  horizon remainder.

No such derivation is supplied here. The RG, data-processing, one-loop, BBN, and asymptotic-safety passages in the earlier skeleton were analogies or leading-order observations, not proofs. In particular, the ideal-radiation identity  $T = (1 - 3w)\rho = 0$  is correct for  $w = 1/3$ , but it does not by itself guarantee sub-percent BBN safety once massive species, trace anomalies, and the background scalar dynamics are included.

The earlier definition

$$\beta_{\Lambda}^{\text{old}} = \left. \frac{d}{d \ln a} \frac{\partial \Phi}{\partial \rho} \right|_{\rho=\rho_*(a)}$$

is identically zero because the bracket is zero for every  $a$ . It cannot yield a non-zero flow. If one merely differentiates the conditional ansatz  $\rho_*^{\text{ansatz}} \propto H^2 M_{\text{Pl}}^2 / L$ ,  $L = \ln(M_{\text{Pl}}/H)$ , the result is

$$\frac{d\rho_*^{\text{ansatz}}}{d \ln a} = \rho_*^{\text{ansatz}} \left( 2 + \frac{1}{L} \right) \frac{d \ln H}{d \ln a}. \quad (19)$$

This is a response of an assumed scaling, not an independently extracted beta function and not evidence for that scaling.

No Wilsonian beta function, fixed-point monotonicity, bound on  $\gamma$  at one loop, or Kullback–Leibler data-processing statement is promoted to a result in this revision. Each requires a dimensionally complete action, normalized probability objects where information inequalities are invoked, and an explicit renormalization prescription.

The conditional scale

$$\Lambda_{\text{eff}}^{\text{ansatz}} \simeq 3.8 \times 10^{-55} \text{ m}^{-2}$$

is about 290 times below the observed value. Accordingly, this paper does not claim a solution of the cosmological-constant or coincidence problems, a DESI-compatible equation of state, or a stable late-time attractor.

## 6.1 Metric $f(R)$ Perturbations

Use the convention

$$ds^2 = -(1 + 2\Psi)dt^2 + a^2(1 - 2\Phi)d\mathbf{x}^2$$

and define  $s(k) = (k^2/a^2)/(k^2/a^2 + m_s^2)$ . For  $F = df/dR \simeq 1$ , the quasistatic linear response has the standard form

$$k^2\Psi = -4\pi G a^2 \bar{\rho} \delta \left(1 + \frac{s}{3}\right), \quad (20)$$

$$k^2\Phi = -4\pi G a^2 \bar{\rho} \delta \left(1 - \frac{s}{3}\right). \quad (21)$$

Thus, with the standard numerator convention,

$$\eta_{\text{std}} = \frac{\Phi}{\Psi} = \frac{1 - s/3}{1 + s/3} \longrightarrow \frac{1}{2} \quad (s \rightarrow 1), \quad (22)$$

while the reciprocal convention tends to 2. The Weyl/lensing combination remains GR-like, apart from the usual  $1/F$  factor:

$$\Sigma \simeq \frac{1}{F} \simeq 1. \quad (23)$$

Consequently,  $\eta = 5/4$  and  $\Sigma = 9/8$  are not predictions of standard metric  $f(R)$  gravity and all forecasts based on those values are withdrawn. The  $4/3$  enhancement applies to the force potential  $\Psi$ , not to a universal free-energy coefficient [13, 14].

Strict convexity of a fixed, environment-independent scalar function  $\Phi_{\text{red}}(\rho)$  excludes a second minimum of that same function. It does not rule out chameleon screening, because the effective scalar potential itself changes with ambient density. Likewise, changing only a positive-definite metric on configuration space cannot move the minimizer of a fixed scalar function:  $\text{grad}_g \Phi = 0$  if and only if  $d\Phi = 0$ .

## 6.2 Source-Bound Local Gate

The 2 July 2026 environment ledger uses

$$|f_R(\text{env})| = |f_{R0}| \left( \frac{R_0}{R_{\text{env}}} \right)^{n+1} \quad (24)$$

together with

$$\epsilon_{\text{th}} = \frac{|f_{R,\text{env}} - f_{R,\text{in}}|}{2\Phi_{\odot}}, \quad f_{R,\text{in}} \simeq 0, \quad \Phi_{\odot} = 2.12 \times 10^{-6}.$$

The Cassini threshold  $\epsilon_{\text{th}} < 2.3 \times 10^{-5}$  requires

$$\frac{|f_R(\text{env})|}{|f_{R0}|} < \frac{2\Phi_{\odot}(2.3 \times 10^{-5})}{|f_{R0}|}. \quad (25)$$

For  $n = 1$ , this gives

$$\frac{R_{\text{env}}}{R_0} \geq 350.787 \quad \text{for } |f_{R0}| = 1.2 \times 10^{-5}, \quad \frac{R_{\text{env}}}{R_0} \geq 101.264 \quad \text{for } |f_{R0}| = 10^{-6}.$$

These threshold calculations are arithmetically correct. No independent local density/curvature model currently supplies the required  $R_{\text{env}}/R_0$ ; the ledger therefore permits no claim upgrade. Neither the exploratory best-fit  $1.2 \times 10^{-5}$  nor the conservative  $10^{-6}$  guide is by itself a Cassini proof. The former is background-proxy compatible only and is not demonstrated Cassini-safe.

The previous exploratory MCMC scan used simplified diagonal summary terms rather than the official Planck/DESI likelihoods and encoded Cassini through a linear hard proxy. Its numerical samples may be retained as a background stress test only; they do not establish model preference, official parameter constraints, or local screening. Table 2 provides a synoptic synthesis of the quantitative parameters, relations, and empirical thresholds governing each sector.

Table 2: Synoptic Parameter, Bound, and Falsification Ledger Across All Sectors of the Framework.

| Sector / Quantity                     | Formal Expression                                                                    | Value / Bound                               | Reference / Gate                                      | Epistemic Status                              |
|---------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| Variational minimizer                 | $\rho_* = \frac{B}{2C} W_0 \left( \frac{2C\rho_{\text{Pl}}}{eB} \right)$             | $\rho_* > 0$ for $B, C > 0$                 | Def. 2.1, Eq. (5)                                     | Proven (strictly convex ground state)         |
| Curvature mapping                     | $\Lambda_{\text{eff}} = 8\pi G\rho_*/c^4$                                            | Dependent on $B/C$                          | Eq. (7)                                               | Exact mapping                                 |
| Cosmological ansatz                   | $\Lambda_{\text{eff}}^{\text{ansatz}} \sim \frac{H_0^2}{c^2 \ln(M_{\text{Pl}}/H_0)}$ | $3.8 \times 10^{-55} \text{ m}^{-2}$        | Planck 2018 ( $1.11 \times 10^{-52} \text{ m}^{-2}$ ) | Conditional (290× deficit; $B, C$ open)       |
| Pöschl–Teller curvature               | $V_{\text{PT}}''(0) = -2V_0/\phi_0^2$                                                | Negative ( $r_+ > 0$ )                      | Eq. (11)                                              | Unstable hilltop (tanh proxy only)            |
| Scalaron mass                         | $m_s^2 = 1/(6\gamma)$                                                                | $\gamma > 0$                                | Eq. (10)                                              | Standard $R^2$ gravity                        |
| Gravitational slip                    | $\eta_{\text{std}} = \frac{1-s/3}{1+s/3} \rightarrow 1/2$                            | Reciprocal $\rightarrow 2$                  | Eq. (22)                                              | Standard metric limit (5/4 withdrawn)         |
| Lensing parameter                     | $\Sigma \simeq 1/F \simeq 1$                                                         | GR-like response                            | Eq. (23)                                              | Standard metric limit (9/8 withdrawn)         |
| Cassini gate ( $1.2 \times 10^{-5}$ ) | $R_{\text{env}}/R_0 \geq 350.787$                                                    | $\epsilon_{\text{th}} < 2.3 \times 10^{-5}$ | Eq. (24)                                              | Arithmetic pass; local profile source-blocked |
| Cassini gate ( $10^{-6}$ )            | $R_{\text{env}}/R_0 \geq 101.264$                                                    | $\epsilon_{\text{th}} < 2.3 \times 10^{-5}$ | Eq. (24)                                              | Conservative guide; no screening proof        |
| Radiation trace                       | $T = (1 - 3w)\rho$                                                                   | $T = 0$ for $w = 1/3$                       | Massless fluid                                        | Exact identity; BBN quantitative gate open    |

## 7 Proof Status and Next Steps

Table 3 formalizes the post-audit status across all seven investigative components, strictly separating proven mathematical results from conditional ansätze and retracted assertions.

The next decisive calculations are: (i) determine dimensionally explicit, renormalized  $B(H)$  and  $C(H)$ ; (ii) replace the hilltop potential or derive a stable scalar history from the full equations; (iii) supply an independent galactic/solar Hu–Sawicki profile whose curvature ratio meets the ledger threshold; and (iv) only then rerun an official cosmological

Table 3: Formal Audit Ledger of Mathematical and Physical Claims After the 16 July 2026 Audit.

| Item                         | Status after the 16 July 2026 audit                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Reduced functional           | Proven: existence, uniqueness, strict convexity, and the Lambert- $W$ minimizer for specified $B, C > 0$ .                           |
| Raw mode sum                 | Invalid as a dimensionally homogeneous functional; motivation only.                                                                  |
| Physical $H_0^2/\ln$ scaling | Open and conditional on explicit RG matching; current arithmetic misses observation by about 290.                                    |
| Pöschl–Teller dynamics       | Hilltop transient; no stable de Sitter attractor and no derived tanh law.                                                            |
| Metric $f(R)$ slip/lensing   | Standard unscreened limits are $\eta_{\text{std}} \rightarrow 1/2$ (or reciprocal 2) and $\Sigma \simeq 1/F$ , not $5/4$ and $9/8$ . |
| Local screening              | Open: threshold ledger correct, but all numerical passes are source-blocked.                                                         |
| Exploratory MCMC             | Background-level proxy only; no official Planck/DESI likelihood and no Cassini-safe classification.                                  |

likelihood. These gates block stronger physical claims and a validated-model release. A corrective transparency edition retains the reduced mathematical result and explicitly records the unfulfilled gates.

## AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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